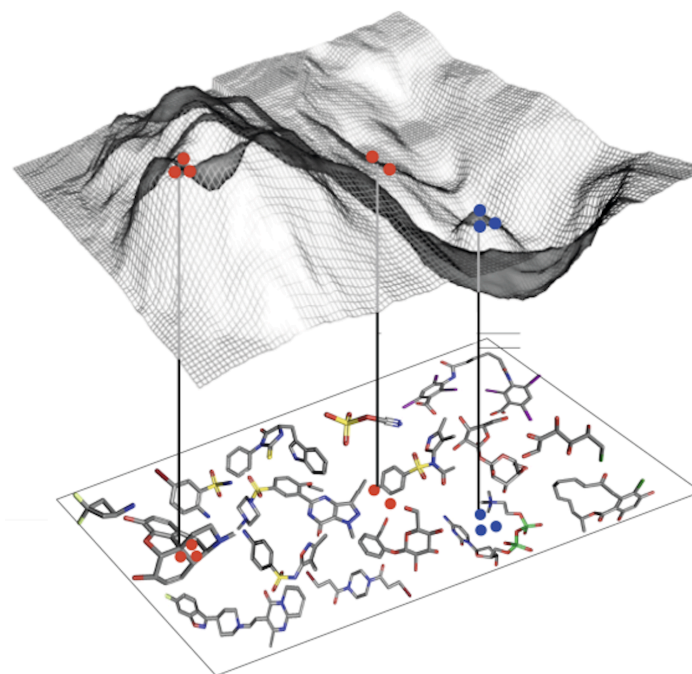


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AI in Drug Discovery

by Vladislav Antipin and Lysandre Terrisse



ABSTRACT

- ❑ Developing a drug = searching in an **extremely large space with noisy objectives**
- ❑ Molecules can be represented as **feature vectors, strings, graphs or 3D signals**
- ❑ **Most of state-of-the-art AI** models are being adapted to chemical data
- ❑ AI is good at **early-stage research**
- ❑ **Biology remains the major bottleneck**
- ❑ One can flip the objective and **design dangerous molecules** on purpose

GLOSSARY

Clinical trial: a type of research that studies new tests and treatments and evaluates their effects on human health outcomes. *WHO*

Chemical space: the vast and theoretically infinite universe of all possible chemical compounds, including both known and hypothetical molecules. *DeepOrigin Computational Chemistry dictionary*

Binding affinity: refers to the strength of interaction between a drug and its target receptor. *ScienceDirect*

ADME: abbreviation for absorption, distribution, metabolism, and excretion, and is mainly used in fields such as pharmacokinetics and pharmacology. *Wikipedia*

Tabular ML: “classical” machine learning models applied to tabular data, e.g. linear models or random forest.

Transformer: a model replacing the recurrent layers most commonly used in encoder-decoder architectures with multi-headed self-attention, allowing to draw global dependencies between input and output. *Attention Is All You Need*

GNN: deep learning models designed to work with graph-structured data, where information is represented as nodes and edges. *GeeksforGeeks*

(3D) CNN: neural network architectures with multiple layers that can learn hierarchical data representations. *NeuralConcept*

VAE: generative models that learn a smooth, probabilistic latent space, allowing them not only to compress and reconstruct data but also to generate entirely new, realistic samples. *GeeksforGeeks*

GAN: class of deep learning architectures with two neural networks (generator and discriminator) competing with each other in the form of a zero-sum game, where one agent's gain is another agent's loss. *Wikipedia*

Normalizing flow models: models providing a general mechanism for defining expressive probability distributions, only requiring the specification of a (usually simple) base distribution and a series of bijective transformations. *Papamakarios et al. 2019*

Diffusion model: neural networks that are trained through deep learning to progressively “diffuse” samples with random noise, then reverse that diffusion process to generate high-quality samples. *IBM*

Reinforcement learning: a type of machine learning process in which autonomous agents learn to make decisions by interacting with their environment. *IBM*

Evolutionary algorithm: an optimization algorithm that has mimicked the biological mechanism such as mutation, recombination, and natural selection to find an optimal design within specific constraints. *ScienceDirect*

Pareto front: the set of non-dominated solutions in multi-objective optimization problems, where no single solution can outperform all others across all objectives. *ScienceDirect*

Hit: a compound which has the desired activity in a compound screen. *Sigma-Aldrich*

Oracle problem: refers to the inability of an agent to know if the information coming from an oracle is authentic and unbiased. *Caldarelli et al. 2025*

Distribution shift: a common situation in machine learning tasks, where the data used for training a model is different from the data the model is applied to in the real world. *Acevedo et al. 2024*

Shortcut learning: a phenomenon in which an AI model learns to solve a task based on spurious correlations present in the data as opposed to features directly related to the task itself. *Ong Ly et al. 2024*

Reality gap: a big mismatch between simulation and reality. *NVIDIA Docs*